THE DUAL OF THE COKERNEL OF AN INJECTIVE ENDOMORPHISM OF A FINITE FREE MODULE VANISHES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Equivalence between the rank of a map and the non-vanishing of its ideal of maximal minors. Lemma \ref{lemma-exact-length-1} and its equivalence of (1) and (2), cited at proof lines 3909-3912. Turns the rank comparison into injectivity of the transpose.

Source reference. Original lemma 0EWY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 18638–18638. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 00MX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 839–839. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a ring. Suppose that $\varphi : R^n \to R^n$ be an injective map of finite free modules of the same rank. Then* $\mathrm{Hom}_R(\mathrm{Coker}(\varphi), R) = 0$.

Proof. Let $\varphi^t : R^n \to R^n$ be the transpose of φ. The lemma claims that φ^t is injective. With notation as in Lemma 00MX we see that the rank of φ^t is n and that $I(\varphi) = I(\varphi^t)$. Thus we conclude by the equivalence of (1) and (2) of the lemma. □

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